

Electric Circuits

Synthetic study fixture for Memo iOS import testing. No personal information.

1. Current, voltage and resistance

Electric current is the rate at which electric charge flows. It is measured in amperes (A).

Voltage is the electrical potential difference between two points, measured in volts (V).

Resistance opposes current and is measured in ohms. A closed circuit is needed for a steady current through a simple battery, wire and resistor circuit.

2. Ohm's law

For an ohmic resistor at constant temperature, voltage equals current times resistance:

$V = I \times R$. With a 12 V supply and a 6 ohm resistor, current is 2 A.

At a fixed voltage, increasing resistance reduces the current.

3. Series and parallel circuits

In a series circuit the same current passes through every component.

Series resistances add: two resistors of 3 ohms and 6 ohms total 9 ohms.

In parallel, each branch has the same voltage and branch currents add.

Two equal 6 ohm resistors in parallel have an equivalent resistance of 3 ohms.

4. Electrical power

Power is the rate of energy transfer. Electrical power can be calculated with $P = V \times I$.

A resistor with 12 V across it and 2 A through it transfers energy at 24 watts.

For classroom experiments use a suitable low-voltage source. Never use wall outlets.

Review questions

What happens to current if resistance doubles while voltage remains fixed?

Why does a break in a simple series circuit stop the current?

Calculate the current through a 4 ohm resistor connected to an 8 V supply.